

Coursework Guideline

July 4, 2026

1 Overview

Mosquitoes are one of the deadliest animals by the number of human deaths they cause, as some mosquito species are major vectors of diseases such as malaria, dengue, and yellow fever. Mosquitoes produce characteristic flight and wingbeat sounds, so we would like to develop a machine learning approach to detect them by sound.

In this coursework, you will develop suitable classification techniques to detect mosquitoes from audio recordings. The key challenge is not only to perform well on the provided datasets, but also to generalise to **out-of-distribution** samples, i.e. recordings drawn from other datasets or captured on different devices, such as mobile phones. You will create slides to introduce the problem, your solutions, and results. You will need to submit a Jupyter notebook by 12:00 noon on Saturday 18th July 2026, with the group presentations held the same day. Any of your experiment figures and/or tables included in your presentation need to be reproducible in the Jupyter notebook that you submit.

2 Datasets

Mosquito data (provided): For mosquitoes, we use an open dataset (<https://zenodo.org/records/4904800>), which provides a large set of multi-species sound samples and their ground truth labels. It includes a multi-part zip of audio files (around 4GB in total) and a CSV file “neurips_2021_zenodo.0_0.1.csv” containing the metadata. Mosquito presence is indicated in Column F (“sound_type”: mosquito, audio, background). More information about the dataset can be found in [1].

OOD test data (self-sourced): To test generalisation, you are expected to assemble a small out-of-distribution test set yourselves, for example mosquito recordings from a different

dataset than the one used for training, or short clips you record yourselves on a mobile phone, together with everyday background recordings. Document clearly where each recording came from and how you labelled it.

Essential task: Build models to detect mosquito presence from audio. Crucially, you must demonstrate how well your models generalise to **out-of-distribution (OOD)** samples, i.e. audio that differs from your training data in its source dataset or recording device. You should assemble a small OOD test set for this purpose, for example recordings taken from a different dataset than the one used for training, or short clips you record yourselves on a mobile phone.

The goal is a **practically useful** detector. Imagine your model deployed in a phone app used in everyday conditions: it should keep good sensitivity to genuine mosquito sounds while keeping false positives low when it hears everyday natural and background sounds (for example wind, traffic, speech, music, or other insects). You should show, with evidence, how your model behaves in such realistic scenarios and how you reduced false alarms without sacrificing detection of the target sounds.

Notes: Present your results in a way that demonstrates how well your model is likely to perform on unseen and out-of-distribution samples, not only on data drawn from the same distribution as your training set.

3 Group allocation

You should form a group of 2 people to complete the coursework. If you want to be a single-person team, this is fine and just let me and Prof Cui know. Please form your groups proactively and finalise them with Prof Cui in good time. Each person will need to explain in the final presentation the individual contribution he/she makes to the completion of the coursework.

4 Deliverables

You should submit the following two files:

1. A presentation slides that describes the problem, literature review, your approach and evaluation results.
2. A Jupyter notebook containing code to train and evaluate your developed algorithms with the data available. The submitted Jupyter notebook needs to include the figures for evaluating performance which you included in your slides.
3. You can optionally include the dataset you collected to support reproducible results, although this is not mandatory.

There will be a group presentation of 7 minutes for each group on Saturday 18th July 2026, on which the evaluation is based. A Chinese presentation is fine, although an English presentation is encouraged with a 5 point bonus added to the coursework score.

5 Coursework slides

It is important to demonstrate your understanding of the subject area, your approach to the coursework and present the outcome. An example of a research presentation can be found here: https://tair-lab.github.io/projects/bdc_eccv/.

A talk on creating effective slides can be found here: <https://www.youtube.com/watch?v=meBXuTIPJQk>.

You should consider including sufficient evidence of your work through the presentation, covering the following areas in the marking criteria.

6 Marking criteria

You will be evaluated based on the following criteria with your **presentation** and **slides**.

- **Introduction to the project:** Provide appropriate background information, and detail the objectives of the project. Literature review should be carried out on relevant topics and how they may relate to your work. You are expected to research and discuss other sources of information, but must show their origins by referencing all sources used. The reference list should be included in the slides. (20/100)
- **Analysis of the problems:** Research and technical issues, challenges, and description of the approach you have developed with justification. Effective graphic illustration will be appreciated. Multiple classification techniques can be designed and implemented. It could be also a multiple versions of the same technique, with different hyperparameters or different ways of training approaches, so you can compare the performance across different models in the slides. (20/100)
- **Data collection efforts:** Discuss the out-of-distribution test set you assembled, including the everyday background recordings you gathered to probe false positives. Give examples of your training and testing audio, document the origin of each source and how you labelled it, and explain how your OOD samples differ from the training data (e.g. different dataset or recording device). (20/100)
- **Implementation and evaluation of the methods:** You need to discuss the system results, present the data you used, how many of them for training and validation in your model, and what are suitable performance metrics etc. (hint: you may want to include not only accuracy here as your performance metrics). You must report performance separately on in-distribution and out-of-distribution samples, and discuss any gap between the two and how you tried to close it. You should also evaluate practical usefulness for a real-world setting such as a phone app: report sensitivity (recall) to genuine mosquito sounds alongside the false positive rate on everyday natural and background sounds, and discuss the trade-off between the two. Figures and/or tables can be useful to present such information, especially when multiple approaches are

developed and comparison of various approaches/choices are conducted. Figures included in the slides need to be reproducible as demonstrated in your Jupyter notebook (otherwise they will not be counted in the scores). (40/100)

- An English presentation is encouraged with a 5 point bonus score. (bonus: 5/100)

References

- [1] Ivan Kiskin et al. HumBugDB: a large-scale acoustic mosquito dataset. In *Proc. Adv. Neur. Inf. Process. Sys. (NeurIPS 2021) Track on Datasets and Benchmarks.*, 2021.